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Local bit plane adjacent neighborhood dissimilarity pattern for medical CT image retrieval

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Abstract

In this article, new feature descriptor local bit plane adjacent neighborhood dissimilarity pattern (LBPANDP) is introduced for CT image retrieval. In the proposed method, the input image is first decomposed into eight binary bit planes. Then in each of the first four most significant bit planes, for each center pixel, the dissimilarity information of its binary neighbors are encoded into a single value. The encoded values of each bit plane is then compared with the intensity of the center pixel and this relationship is finally encoded to devise the LBPANDP descriptor. The proposed descriptor is highly discriminative and considers only the first four most significant bit planes for encoding, which greatly reduces the dimension of the feature vector. Retrieval performance of LBPANDP is investigated on two CT image databases and the results show a significant improvement in retrieval efficiency by LBPANDP over many recent techniques.

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Keywords: local pattern; image retrieval; bit plane slicing; coarse textures; dissimilarity

1. Introduction

Computerized tomography (CT) images are an important class of medical images which are used by doctors and clinicians for medical diagnosis, clinical decision making and educational purposes. Now a days with tremendous growth in imaging technologies, CT images produced in hospitals are piling up on a regular basis, resulting into huge image libraries. Proper indexing and classification of these ever increasing volume of image data is important for their pertinent utilization. Therefore, efficient image retrieval systems are garnering high demand so that images can be effectively retrieved and used when they are required. Since human perception based indexing is time consuming and often inaccurate too, content based image retrieval (CBIR) systems are now a days mostly assigned to do the task of indexing and classification of the images. In CBIR systems, first the visual contents of the images are analyzed and then classification is done using some suitable similarity measure techniques such as 'Euclidian', 'chi-square', ' d_1 ' etc. wherein the distance between the contents of the images are computed. The feature descriptors are used to analyze and encode the contents of an image into a low dimensional data array which is termed as feature vector. Approaches including intensity histogram based methods [1], physiological kinetic features [2], wavelet

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transformbased approaches [3] and deep learning based methods [4] etc. are espied in the literature for content based image retrieval of bio medical images. Recently, local pattern based approaches are acquiring substantial attention among the researchers for bio medical image retrieval applications. Besides being efficient for feature extraction, one principal advantage of local pattern based approaches is that they are usually computationally simpler as compared to the aforesaid approaches. The popular Local Binary Pattern (LBP) was proposed by Ojala et al. in [5] which analyzes the image texture by thresholding the local neighboring pixels with the center pixel and then encoding them into a pattern value in the local circular neighborhoods of the image. LBP is used for biomedical image retrieval in [6] by Unay et al. Over the years several new variants [7,8,9,10,11,12] are proposed to improve the retrieval efficiency of LBP in medical CBIR.

In [13,14], Dubey et al introduced the concept of bit plane based local pattern for feature extraction and retrieval of medical CT images. In this type of encoding, first an image is divided into a number of binary bit plane slices and then local patterns are calculated from these slices for feature extraction. In [13] local bit plane decoded pattern (LBPD) was proposed where for a given center pixel the bit plane transformed values are calculated by applying weights on the binary contents of each bit plane of the neighboring pixel and then the weighted values are summed up to a value. The bit plane transformed values of each of the bit planes are then compared with the intensity of the center pixel to obtain the LBPD feature map. Local bit plane dissimilarity pattern (LBDisP) on the other hand, evaluates the dissimilarity between the local neighboring pixels and the center pixel in each bit plane [14] to construct the feature map.

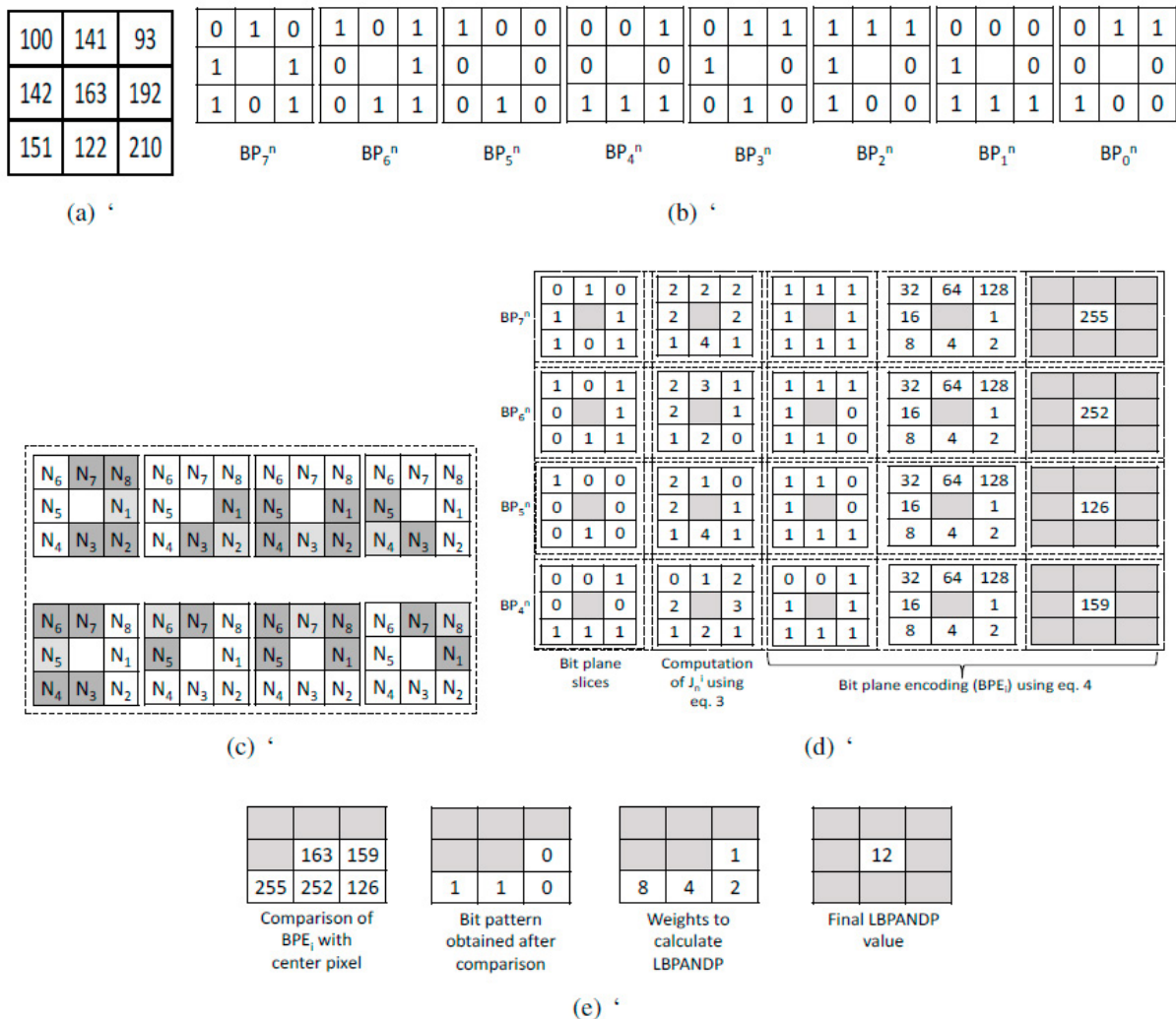


Fig. 1. Example of proposed LBPDNDP computation (a) sample neighborhood; (b) Bit plane slices; (c) corresponding adjacent neighborhoods of a given neighbor; (d) Encoding of the MSB planes BP_i^n ($i=4,5,6,7$) considering adjacent neighbors resulting into bit plane encoded BPE_i values; (e) LBPDNDP computation.

Usually, each bit plane provides unique texture information about the image. Among the bit plane slices, the lower bit planes contain the fine texture information of the image and the higher bit planes contain the coarse textures of the image [15]. The highest bit plane or the most significant bit (MSB) plane usually contain the most meaningful visual information. On the other hand, the lowest or the first bit plane contain relatively much less information [16]. One major advantage of local bit plane encoding is that it accentuates the contribution of each of the binary bits to the overall appearance of the image. Besides textures of different types can be analyzed separately in bit plane encoding.

The approaches discussed in [13] and [14] showed the effectiveness of bit plane encoding in retrieval of bio medical images. However, challenges still lie in extracting more comprehensive features from bit planes to facilitate more accurate and faster retrieval. Here, we attempt to investigate and acquire that information from bit planes which could enhance the discrimination capability of the extracted features.

In local pattern based encoding usually the local relationship between the neighboring pixels with respect to the center pixel is considered to extract the local texture features. Unlike [5,13,14], we first locally encode the dissimilarity information between the neighbors in each bit plane. This extracted dissimilarity information of neighbors in each bit plane is then encoded to a single value ranging between 0-255. Finally, the relationship between the dissimilarity information of each bit plane and the intensity of center pixel is computed using LBP to construct LBPANDP descriptor. Only four most significant bit planes are considered in LBPANDP descriptor which greatly reduces the feature vector dimensions.

The rest of the paper is organized as follows: In section 2 proposed LBPANDP approach is elaborated. The experimental observations are shown in section 3. The paper is concluded in section 4.

2. Proposed approach: local bit plane adjacent neighborhood dissimilarity pattern (LBPANDP)

The proposed approach can be summarized into following steps: Bit plane slicing, bit plane encoding considering adjacent neighbors, LBPANDP computation and Construction of feature vector. An example computation of LBPANDP approach is shown in Figure. 1.

2.1. Bit plane slicing

Here for a given image I the intensity of a pixel element $I(x,y)$ at coordinate (x,y) is divided into B number of bit plane slices (here B is taken as 8) as

$$I(x,y) = \sum_{i=0}^{B-1} BP_i(x,y) \times 2^i \quad (1)$$

where, $BP_i(x,y)$ is the binary form of $I(x,y)$ in i^{th} bit plane (where $i \in [0,7]$). The bit planes of the image I can be visualized as individual images as shown in Figure 2 where the bit planes of a sample image are shown. The higher bit planes usually contain coarser details of the image texture, while the lower bit planes contain the finer details.

For LBPANDP encoding, a local neighborhood of radius R (here $R=1$) is considered around a center pixel of intensity $I(x,y)$ at coordinate (x,y) . The neighborhood radius $R=1$ is considered in order to exploit more local statistics as this provides closer neighborhood relationship between center and its neighbors. The binary form of intensities of the neighbor pixels $I_n^R(x,y)$ (where $n \in [1,8]$) around the center pixel are realized in 8 binary bit planes as

$$I_n^R(x,y) = \sum_{i=0}^7 BP_{i,n}^R(x,y) \times 2^i \quad (2)$$

where $BP_{i,n}^R(x,y)$ is the binary bit of n^{th} neighbor of intensity $I_n^R(x,y)$ around the center pixel in i^{th} bit plane. Figure 1(a-b) shows a sample neighborhood and its bit plane slices. It should be noted that the size of image has no effect on the process of bit plane conversion as the number of bit planes depends only on the bit depth of the image.

2.2. Bit plane encoding considering dissimilarity information of adjacent neighbors

As the most significant bit (MSB) planes contain more significant texture information of the image, in LBPANDP encoding

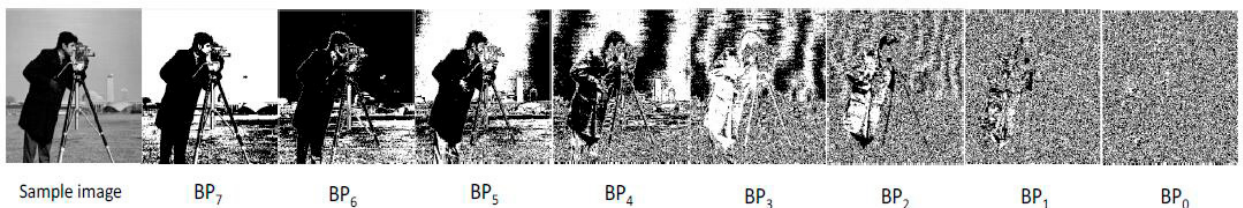


Fig. 2. An illustration of a sample image and its bit plane slices (BP_i).

we have considered only four MSB planes i.e. BP_7, BP_6, BP_5, BP_4 for encoding. It is observed that the LBPANDP descriptor with only four MSB planes still carries highly discriminative information of the image which can be described by much less feature vector dimensions.

In this step all the four considered bit planes ($BP_7, \dots, BP_4$) are encoded separately wherein a binary bit $BP_i(x, y)$ is encoded in to a bit plane encoded value $BPE_i(x, y)$ ($i \in [4, 5, 6, 7]$). For encoding a binary bit $BP_i(x, y)$ at coordinate (x, y) in i^{th} bit plane neighboring binary bits $BP_{i,n}^1$ (where $n \in [1, 8]$) in the local neighborhood of radius 1 around $BP_i(x, y)$ are considered. For simplicity we represent $BP_{i,n}^1$ as N_n in the next part of discussion. For a given neighbor bit N_n its adjacent neighbors are selected as shown in Figure 1(c). It can be observed from Figure 1(c) that for each diagonal neighbors (N_n where $n=2, 4, 6, 8$), there are two adjacent neighbors, while for each horizontal and vertical neighbors (N_n where $n=1, 3, 5, 7$), there are four adjacent neighbors. The adjacent neighbors for a given neighbor pixel N_n are represented as AN_n^t , (where $t \in [1, \dots, N_A^n]$ and N_A^n is the total number of adjacent neighbors). Therefore, for $n=1, 3, 5, 7$, $N_A^n = 4$ and for $n=2, 4, 6, 8$, $N_A^n = 2$.

Now the dissimilarity between the neighbor bit of interest (N_n) and its adjacent neighbors (AN_n^t) are computed as

$$J_n^i = \sum_{t=1}^{N_A^n} Ds(N_n, AN_n^t); \text{ where } Ds(a, b) = \begin{cases} 1; & \text{if } a \neq b \\ 0; & \text{else} \end{cases} \quad (3)$$

J_n^i gives the number of dissimilar adjacent neighbor bits for a given neighbor bit N_n in i^{th} bit plane. Since $N_A^n = 4$ for $n=1, 3, 5$ and 7, values of J_n at these positions are in the range $[0, 4]$. Similarly, for $n=2, 4, 6$ and 8, values of J_n are in the range $[0, 2]$ since $N_A^n = 2$ at these positions.

Now, $BP_i(x, y)$ is encoded into a bit plane encoded (BPE) value by thresholding J_n^i values with a threshold Th as follows:

$$BPE_i(x, y) = \sum_{n=1}^8 2^{n-1} \times f_i(J_n^i, Th); \text{ where } f_i(p, q) = \begin{cases} 1, & \text{if } p \geq q \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

where, the threshold Th is taken as $\frac{N_A^n}{2}$; $BPE_i(x, y)$ is the bit plane encoded value in i^{th} bit plane at coordinate (x, y) . Since we considered only four bit planes ($i=4, 5, 6$ and 7) for encoding, we get four BPE_i values at a given coordinate. These values are in the range of $[0-255]$ as shown in Figure 1(d).

2.3. LBPANDP computation

Figure 1(e-h) demonstrate computation of final LBPANDP. In this step, the relationship between $BPE_i(x, y)$ values and the intensity of the center pixel i.e. $I(x, y)$ is used to obtain the pattern value as

$$LBPANDP(x, y) = \sum_{i=4}^7 2^{i-4} \times f_i(BPE_i(x, y), I(x, y)) \quad (5)$$

The resultant LBPANDP pattern map of the image I contains $2^4 = 16$ distinct pattern values in the range 0-15.

2.4. Construction of feature vector

In this step, the LBPANDP pattern map is first divided into a number of non-overlapping sub maps of same size. From each of these maps, histograms are computed separately using (5) and (6). At last all these histograms are concatenated together to construct the final feature vector.

$$X(\zeta) = \sum_{x=1}^{X_1} \sum_{y=1}^{X_2} H(LBPANDP(x, y), \zeta); \text{ where } H(p, q) = \begin{cases} 1, & \text{if } p = q \\ 0, & \text{otherwise} \end{cases} \quad (6)$$

$X_1 \times X_2$ is the dimension of each block.

3. Experimental results

To validate the retrieval performance of the proposed descriptor experiments are carried out on two publicly available CT images databases, namely NEMA [19] and TCIA [20]. The image details of these databases are provided in Table 1 and 2. The proposed approach is compared with seven well known approaches which are LBP (Dim: 256) [5], LDEP (Dim: 24) [9], CSLBCoP (Dim: 1024) [10], LBDP (Dim: 256) [13], LBDiSP (Dim: 256) [14], LWP (Dim: 256) [11] and CoCSLBP (Dim: 64) [17]. The retrieval performances of proposed method and other methods are compared using two standard parameters average retrieval precision (ARP) and average retrieval recall (ARR) [17,18]. In the experiments, every image in the databases are taken as query and against each query a set of images are retrieved from the database. The total number of relevant images retrieved for

all such queries are then used to calculate the ARP and ARR values. We considered 4 non overlapping blocks to calculate histograms in the proposed approach.

Table 1. Summary of National Electrical Manufacturers Association (NEMA) database.

Database name	Image details (data set name: number of images)	Total images
NEMA	CT0001: 36, CT0003: 18, CT0003: 36, CT0057: 37, CT0057: 41, CT0060: 30, CT0060: 23, CT0080: 70, CT0083: 24	315

Table 2. Summary of The Cancer Imaging Archive (TCIA) database.

Database name	Image details (subject ID: number of images)	Total images
TCIA	1.3.6.1.4.1.9328.50.5.0001: 59, TCGA-CS-6186: 25, TCGA-K7-A6G5: 32, PANCREAS 001: 40, PANCREAS 003: 31, C3N-02012: 42, AMC-027: 43, BreastDx-01-0075: 25, TCGA-ZF-AA5P: 20, TCGA-FI-A2F8: 36, TCGA-DE-A4MC: 44, C3N-02275: 73, 1.3.6.1.4.1.9328.50.4.0001: 57,69,63,PANCREAS 0003: 37	696

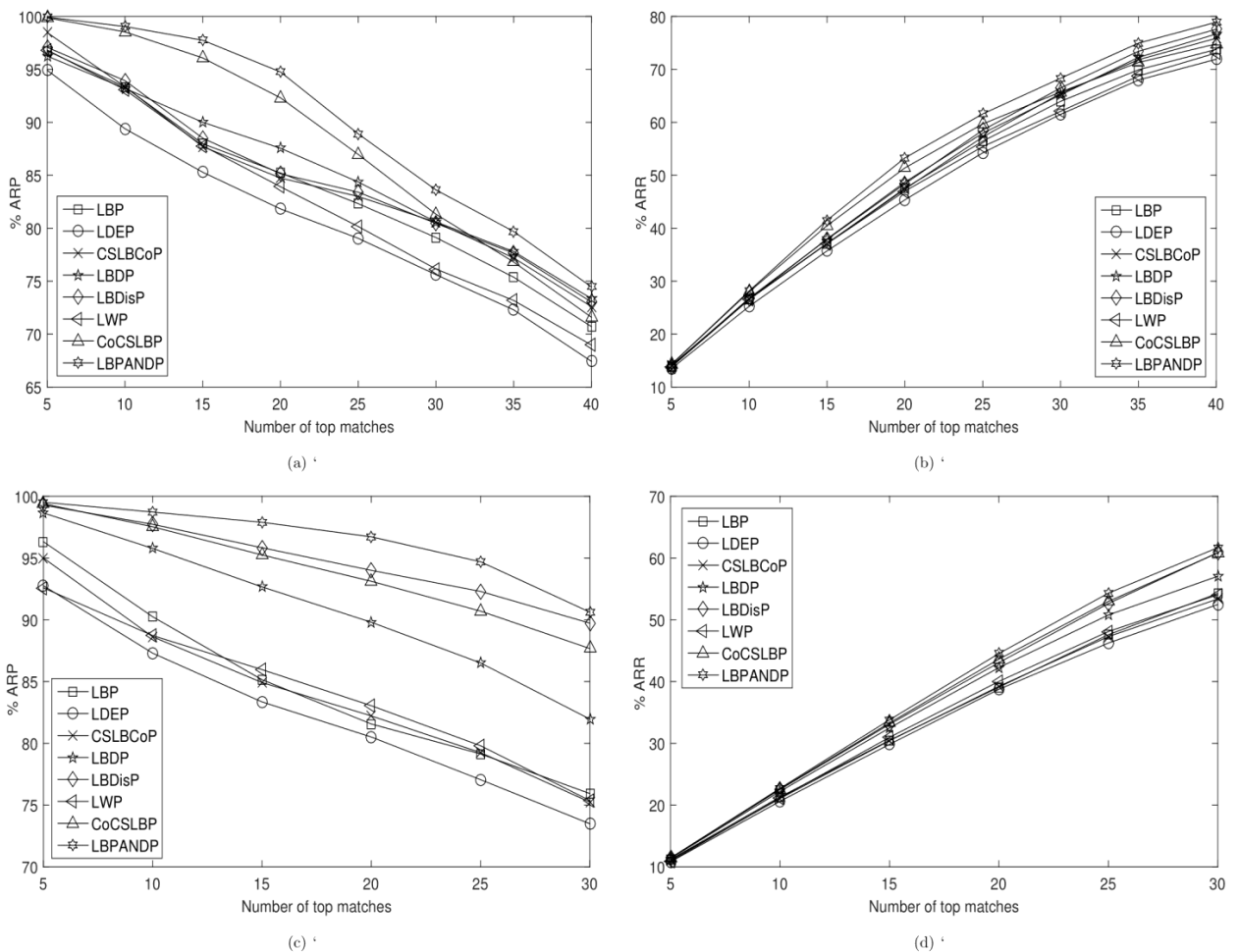


Fig. 3. Retrieval results comparison of LBPANDP and other methods using % ARP and % ARR for (a-b) NEMA database and (c-d) TCIA database.

3.1. Experiment with NEMA database

The first experiment is conducted on the CT images collected from NEMA database [19]. The case details of the CT images considered in our database is provided in Table 1. A total of 315 CT images of 9 different classes are considered. The number of CT images in each class are 36, 18, 36, 37, 41, 30, 23, 70 and 24 respectively. To evaluate the retrieval performance, ARP and ARR values are measured for 5, 10, 15, ... 40 numbers of top matches. % ARP and % ARR graphs of different approaches

are shown in Figure 3(a-b). In Table 3, % ARP and % ARR values of different approaches are shown for 40 number of top matches. From the plots and table, we can infer following points:

1. The proposed method consistently shows better performance than the other bit plane based methods LBDP and LBDiSP in all the top matches.
2. Although the proposed method LBPANDP performs very close to CoCSLBP method in lower number of top matches, but with the increase in number of top matches LBPANDP consistently outperforms the CoCSLBP by a decent margin.
3. The proposed method consistently outperforms the high dimensional CSLBCoP method in all the top matches.
4. For 40 top matches, LBPANDP outperforms LBP, LDEP, CSLBCoP, LBDP, LBDiSP, LWP and CoCSLBP methods by 5.30 %, 10.43 %, 2.57 %, 1.50 %, 1.96 %, 7.96 % and 4.00 % respectively, in terms of % ARP.

3.2. Experiment with TCIA database

For the second experiment, TCIA-CT image database [20] is considered. Table 2 shows the case details of the CT images that are considered in our experiments. The collected CT images are of various portions of the body from the TCIA database. A total of 696 images of 16 different classes are considered in our experiment. The number of images in each class are 59, 25, 32, 40, 31, 42, 43, 25, 20, 36, 44, 73, 57, 69, 63 and 37 respectively. To evaluate the retrieval performance, ARP and ARR values are computed for 5, 10, 15, ... 30 number of top matches and the plots of ARP and ARR for different approaches are shown in Figure 3(c-d). Following points can be drawn from these plots:

1. The proposed method consistently outperforms the other bit plane based methods LBDP and LBDiSP in all the top matches.
2. LBDiSP performs very close to the proposed approach, which is followed by CoCSLBP and LBDP approaches.
3. For 30 top matches, LBPANDP improves upon % ARP of LBP, LDEP, CSLBCoP, LBDP, LBDiSP, LWP and CoCSLBP methods by 19.34 %, 23.31 %, 20.44 %, 10.57 %, 0.99 %, 20.24 % and 3.36 % respectively.

We investigated the retrieval performance of LBPANDP in terms of % ARP using four similarity distance measurement methods which are Manhattan, Euclidean, d_1 and chi-square [17]. From the experiments it is observed that LBPANDP performs best with chi-square distance measure in NEMA database, while in TCIA database it performs best with d_1 distance measure.

The % ARP and % ARR values of different methods for both the databases and their respective feature dimensions are shown in Table 3. From the table it is observed that the proposed LBPANDP outperforms all other approaches with relatively less number of features. The feature dimension of LBPANDP is 2.67 times higher than LDEP and is same as CoCSLBP, while its dimension is 0.25 times of LBDP, LBDiSP and LWP approaches and 0.0625 times that of CSLBCoP. However, LBPANDP still exhibits superior retrieval efficiency compared to all other approaches.

Table 3. % ARP and % ARR of different methods for NEMA (for 40 top matches) and TCIA (for 30 top matches) databases along with their respective feature dimension.

Database	Performance parameters	Retrieval Methods						
		LBP	LDEP	CSLBCoP	LBDP	LBDiSP	LWP	CoCSLBP
NEMA	% ARP	70.71	67.43	72.60	73.36	73.03	68.97	71.62
	% ARR	73.78	71.92	75.88	76.69	77.59	73.12	74.82
TCIA	% ARP	75.93	73.49	75.24	81.96	89.73	75.36	87.67
	% ARR	54.27	52.48	53.38	57.06	60.88	54.04	60.84
Feature Dimension		256	24	1024	256	256	256	64

4. Conclusion

In this paper, an efficient feature descriptor LBPANDP is introduced, where the dissimilarity relationship of neighboring pixels in each bit plane is first captured and then encoded to a value, which is then used to find a relationship with the intensity of center pixel to form the LBPANDP descriptor. In contrast, the recent LBDiSP method simply captures the relationship between the neighboring pixels and the center pixel in each bit plane. Therefore, the LBPANDP descriptor captures additional information compared to LBDiSP descriptor which makes it highly discriminative and shows encouraging results with just first four most significant bit planes which further reduces the size of the feature vector. Experimental results after being investigated on two popular CT image databases validate the retrieval superiority of LBPANDP over some of the recent methods.

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